

NEW CENTURY, KS, USA

WMO: 724475

Lat: 38.832N

Lon: 94.890W

Elev: 1087

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 93909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	9.1	-4.4	4.5	6.5	-0.2	5.7	12.0	27.7	39.1	25.8	36.9	8.9	330	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	97.2	75.7	93.5	75.5	90.8	75.0	79.1	90.8	77.8	89.3	76.6	87.4	11.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	138.9	85.5	74.4	134.0	84.3	73.1	127.9	82.8	43.3	91.1	42.0	88.8	40.7	87.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.3	22.7	19.9	-3.5	101.4	4.9	3.4	-7.1	103.9	-9.9	105.8	-12.7	107.7	-16.3	110.2		
			-4.3	81.2	4.7	1.5	-7.6	82.3	-10.4	83.2	-13.0	84.0	-16.4	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	30.5	34.2	44.9	55.4	65.3	74.5	79.1	77.6	69.6	56.8	44.8	33.6
	DBStd	19.34	11.31	11.54	11.44	9.26	8.03	6.09	5.62	5.88	7.51	9.29	10.15	10.82
	HDD50	2118	607	454	235	50	3	0	0	0	0	40	214	515
	HDD65	4873	1069	862	628	306	99	7	0	1	38	283	607	973
	CDD50	4175	4	12	75	212	476	735	901	856	587	252	58	7
	CDD65	1455	0	0	4	19	107	292	436	392	175	29	1	0
	CDH74	14421	0	1	28	181	902	2793	4779	3954	1545	232	6	0
	CDH80	6158	0	0	2	29	246	1101	2293	1861	572	54	0	0
	WSAvg	9.8	10.5	10.7	11.4	11.7	9.9	9.4	8.1	7.8	8.4	9.5	10.3	10.3
	PrecAvg	39.8	1.1	1.6	2.4	4.1	5.6	5.5	4.1	4.4	4.1	3.0	2.4	1.6
Precipitation	PrecMax	56.4	3.1	3.9	5.9	9.0	12.2	9.5	14.6	10.0	11.2	7.4	7.4	4.4
	PrecMin	24.3	0.1	0.0	0.9	0.7	1.4	1.3	0.8	1.0	0.5	0.2	0.2	0.0
	PrecStd	7.2	0.7	1.0	1.3	1.8	2.8	2.5	2.9	2.4	2.4	1.9	1.8	1.0
	PrecStd	7.2	0.7	1.0	1.3	1.8	2.8	2.5	2.9	2.4	2.4	1.9	1.8	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.7	70.2	77.9	83.9	89.9	96.6	101.3	101.6	96.5	85.4	74.4	65.7
		MCWB	51.1	54.8	61.9	65.9	72.4	75.7	74.0	73.6	73.8	69.6	61.4	58.1
	2%	DB	57.5	63.3	72.5	79.1	85.7	92.1	97.4	97.2	90.1	80.3	69.9	60.3
		MCWB	48.6	53.5	59.4	63.7	70.2	75.1	76.1	75.5	73.2	65.8	59.3	53.6
	5%	DB	52.4	57.2	68.0	75.2	82.4	89.7	94.2	93.1	86.5	75.3	64.6	54.4
		MCWB	44.2	48.6	56.8	61.9	69.1	74.6	76.2	75.1	72.7	63.7	55.5	47.8
	10%	DB	46.5	52.0	63.5	71.7	79.4	86.5	90.9	90.0	82.4	72.0	60.7	49.6
		MCWB	40.2	44.6	54.7	60.6	67.3	73.2	75.9	74.6	69.9	62.8	53.2	43.9
	0.4%	WB	56.5	59.4	65.5	68.8	75.4	79.6	81.4	80.3	77.4	73.0	64.9	60.4
		MCDB	60.1	65.0	74.6	80.2	85.2	90.3	92.2	91.3	87.7	82.0	69.6	64.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.4	54.5	62.2	66.7	73.1	77.5	79.5	78.4	75.4	70.1	60.5	54.5
		MCDB	55.8	61.4	69.2	75.4	82.3	89.0	91.3	90.4	85.6	76.5	66.5	58.3
	5%	WB	44.8	49.8	59.2	64.2	71.0	76.0	78.2	77.1	73.9	67.1	57.7	48.6
		MCDB	51.5	57.6	66.2	72.1	79.7	86.4	89.6	88.7	83.5	72.7	62.7	53.3
	10%	WB	40.6	44.5	54.9	61.8	68.9	74.5	77.1	75.5	72.1	63.8	54.1	44.1
		MCDB	47.0	51.1	62.1	70.3	77.1	84.0	87.9	86.3	80.7	68.8	59.9	49.1
Mean Daily Temperature Range	5% DB	MDBR	18.3	19.0	20.3	20.8	19.6	19.1	19.7	20.3	20.8	20.4	19.5	17.6
		MCDBR	26.4	27.8	26.6	25.3	22.6	21.6	23.3	24.7	23.2	23.9	24.7	24.6
		MCWBR	18.2	18.9	15.8	14.6	11.2	9.0	7.9	8.5	10.0	12.6	15.8	17.9
	5% WB	MCDBR	24.3	25.4	21.2	21.3	20.1	19.6	20.0	21.1	20.1	19.1	21.2	21.9
		MCWBR	18.0	18.7	14.1	14.3	11.3	9.9	8.8	9.1	9.8	12.5	15.6	17.9
Clear-Sky Solar Irradiance	taub		0.290	0.302	0.332	0.381	0.421	0.426	0.433	0.433	0.384	0.340	0.303	0.286
	taud		2.479	2.453	2.426	2.308	2.251	2.268	2.275	2.277	2.366	2.477	2.523	2.515
	Ebn at Noon		284	294	294	284	273	271	268	265	273	276	276	277
	Edh at Noon		25	29	33	40	43	43	42	41	35	29	24	23
All-Sky Solar Radiation	RadAvg		712	956	1241	1579	1810	2058	2047	1819	1522	1093	774	615
	RadStd		74	109	105	101	161	151	129	98	103	124	75	56

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page